

DFD Level-0

External: Student -> [Library System] -> External: Admin

Processes: Search Books, Issue/Return, Manage Inventory

Data Stores: Books, Users, Transactions

DFD Level-1 (Issue Book)

Process: Issue Book

Inputs: Request from Student

Checks: Availability (Books store) -> Update Transactions store

ER Diagram (Conceptual)

Entities: Student(StudentID, Name, Email), Book(BookID, Title, Author, ISBN), Transaction(TransID, Date, DueDate)

Relationships: Student -< Transaction >- Book (many-to-many resolved with Transaction)

Use Case Diagram (Core Actors)

Actors: Student, Librarian(Admin)

Use Cases: Search Book, Issue Book, Return Book, Reserve Book, Manage Inventory, Generate Reports

Sequence Diagram (Issue Book)

Lifelines: Student -> System -> Inventory DB -> Transaction DB -> Notification

Steps: Student requests issue -> System checks inventory -> Creates transaction -> Sends notification

Activity Diagram (Search & Issue)

Start -> Enter Search -> Select Book -> If available -> Issue (confirm) -> Update records -> End

Else -> Show waitlist/reserve -> End

Class Diagram (Simplified)

Classes: Student {studentID, name, borrow() }

Book {bookID, title, isAvailable(), reserve() }

Transaction {transID, issueDate, dueDate, returnDate }